Analogue programming

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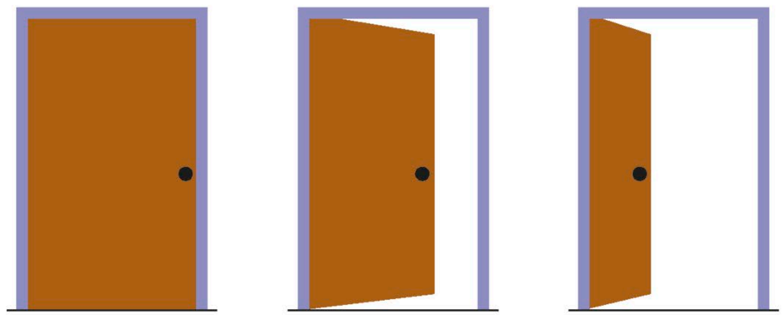
While digital programs work with discrete data formed by 0s and 1s, analogue programs can handle values between these two extremes. They both have a unique approach to programming.

Digital vs analogue data

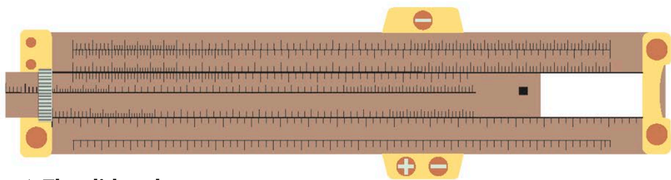
Digital data is limited to specific values. It gives answers in yes or no. Analogue data, on the other hand, gives precise and detailed answers.



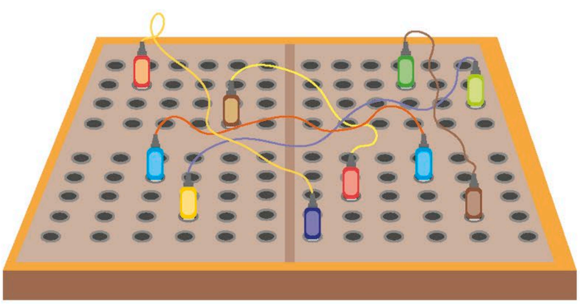
▷ **Limited or precise answers**
Digital data answers questions like, "Is the door open?" with the answer being only "yes" or "no". Analogue data can describe any of the points in between – and so can be used to answer the more accurate question, "How open is the door?".

Analogue computers

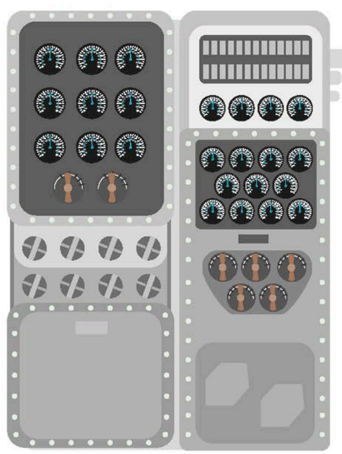
An analogue computer stores and processes data by using physical quantities, such as weight, length, or voltage, as opposed to a digital computer, which stores data as binary code on its hard drive. While digital computers are limited to two values (0 and 1), every single unit of analogue data can give precise answers.



△ **The slide rule**
This mechanical analogue computer was invented in the 1600s. The middle section of the ruler could be slid out to work out mathematical functions by reading the numbers on the scale.



△ **Plugboard**
In analogue computing, there is no concept of software. Programs are created by connecting base circuits using plugboards.



◁ **The Torpedo Data Computer (TDC)**
Used by American submarines during WW II, the TDC was an electromechanical computer that was able to work out the complex mathematics behind firing a torpedo at a moving target, such as a ship.